

EDUCATION

McGill University, Schulich School of Music

Ph.D. in Music Technology

Sep. 2022 - Present

Montreal, Canada

Supervisors: Prof. Marcelo M. Wanderley and Assoc. Prof. Isabelle Cossette

Shanghai Jiao Tong University, USC-SJTU Institute of Cultural and Creative Industry

M.A. in Cultural & Creative Industry Management

Sep. 2019 - Mar. 2022

Shanghai, China

Advisors: Asst. Prof. Gus Xia and Asst. Prof. Zhixin Xu

Peking University, National School of Development

Bachelor in Economics, Double Degree

Sep. 2017 - Jun. 2019

Beijing, China

Beijing University of Posts and Telecommunications, School of Digital Media

B.E. in Digital Media Technology

Sep. 2015 - Jun. 2019

Beijing, China

RESEARCH INTERESTS

My research interests center around music and human-computer interaction, with a particular focus on musical haptics and embodied interaction. By researching music, I aim to inspire the broader HCI field with new paradigms of multimodal interaction, demonstrating how sensing, haptics, and AI-driven modeling of embodied musical practices can inform the design of more intuitive, expressive, and socially connected interactive systems.

PEER-REVIEWED PUBLICATIONS

Journal Articles

- [J.1] **Ziyue Piao**, Christian Frisson, Bavo Van Kerrebroeck, and Marcelo M. Wanderley, "**Assessing the Impact of Force Feedback in Musical Knobs on Performance and User Experience**", *Actuators*, 2024.

Conference Full Papers

- [C.3] **Ziyue Piao**, Yohei Wada, Isabelle Cossette, Marcelo M. Wanderley, and Akira Maezawa, "**Exploring Breathing-Music Coupling: Using the Breathing Mirror for Somatic Reflection in Piano Performance**", *International Conference on New Interfaces for Musical Expression (NIME)*, London, UK, 2026.
- [C.2] Leon Lu, Chase Crispin, **Ziyue Piao**, Aino Eze-Anyanwu, and Audrey Girouard, "**Project TapTap: A Longitudinal Study Exploring Non-Verbal Communication through Vibration Signals Between Teachers and Blind or Low Vision Music Learners**", *ACM International Conference on Human Factors in Computing Systems (CHI)*, Yokohama, Japan, 2025
- [C.1] **Ziyue Piao**, Marcelo M. Wanderley, and Felipe Verdugo, "**MappEMG: Enhancing Music Pedagogy by Mapping Electromyography to Multimodal Feedback**", *International Conference on ArtsIT, Interactivity and Game Creation (ArtsIT)*, Campinas, Brazil, 2023.

Conference Short Papers

Note: All types of publications except full papers

- [P.5] **Ziyue Piao**, Isabelle Cossette, and Marcelo M. Wanderley, "**Reconciling Kinesthetic Mismatches: A Somatic Alignment Mindset for Musical Body Transformation**", *ACM International Conference on Human Factors in Computing Systems (CHI)*, Barcelona, Spain, 2026.
- [P.4] **Ziyue Piao** and Kanyu Chen, "**Respiratory Musical Interfaces (RMIs) in NIME**", *International Conference on New Interfaces for Musical Expression (NIME)*, London, UK, 2026.
- [P.3] **Ziyue Piao**, "**Exploring Haptics to Augment Kinesthetic Learning in Singing**", *ACM International Conference on Tangible, Embedded, and Embodied Interaction (TEI)*, Bordeaux, France, 2025.
- [P.2] **Ziyue Piao** and Gus Xia, "**Sensing the Breath: A Multimodal Singing Tutoring Interface with Breath Guidance**", *International Conference on New Interfaces for Musical Expression (NIME)*, Online, 2022.
- [P.1] Yinmiao Li, **Ziyue Piao**, and Gus Xia, "**A Wearable Haptic Interface for Breath Guidance in Vocal Training**", *International Conference on New Interfaces for Musical Expression (NIME)*, Shanghai, China, 2021.

Yamaha Corporation

Research Intern, Mentors: Dr. Akira Maezawa and Yohei Wada

Sep. 2025 - Present

Hamamatsu, Japan

Breathing Analysis with Sensing Wearables for Piano Performance:

- Led an in-lab study with seven pianists, capturing synchronized piano performance and respiration data through Yamaha GummiStra wearable breathing belts.
- Built an annotated visual interface that layers multimodal breathing and performance signals to streamline note taking and playback.
- Synthesized qualitative interview findings with quantitative breathing and performance metrics to surface design insights.
- Prototyped and iteratively refined the corset with GummiStra sensor to measure thoracic expansion during music performances.

Sony Computer Science Laboratories

Research Intern, Mentors: Dr. Shinichi Furuya, Asst. Prof. Jun Nishida (University of Maryland)

Jun. 2025 - Sep. 2025

Tokyo, Japan

ScapuLearn:

- Developed a machine-learning pipeline that process shoulder-mounted IMU data to estimate finger force, loudness, keyboard register, and scapular posture during piano pieces.
- Analyzed shoulder's rotation and acceleration features and trained gradient-boosted models to infer finger-level dynamics from minimally intrusive upper-body sensing.
- Validated the approach through multi-stage user studies with twenty pianists, demonstrating how wearable sensing plus ML can support performance analysis and pedagogy.

Input Devices and Music Interaction Lab & Music Performance and Body Lab, McGill University

Research Assistant, Advisors: Prof. Marcelo M. Wanderley and Assoc. Prof. Isabelle Cossette

Sep. 2022 - Present

Montreal, Canada

Challenges and Design Opportunities in Teaching Singing Breath Control:

- Led focus groups with eleven teachers and students to map kinesthetic pain points in teaching breath control and prioritize assistive technology concepts.

Haptic Music Editing Interface for Pitch Modulation:

- Built a web interface that streams TorqueTuner haptic data into a pitch-modulation synthesizer for real-time editing.
- Conducted controlled studies on how various force-feedback modes affect fine-grained vibrato control.
- Analyzed both pitch-contour performance data and qualitative feedback via Likert-scale ratings and interviews.

Creative Interactions Lab, Carleton University

Visiting Researcher, Advisors: Prof. Audrey Girouard, Leon Lu

May. 2024 - Aug. 2024

Ottawa, Canada

Wearable Haptics for Music Accessibility:

- Performed thematic analysis of a ten-week study with twelve music teachers and visually impaired students to document how a wearable vibrotactile tool expands music-accessibility strategies.

Laboratory of Simulation and Movement Modelling, University of Montreal

Research Intern, Advisors: Prof. Felipe Verdugo and Prof. Marcelo M. Wanderley

Sep. 2022 - Aug. 2023

Montreal, Canada

MappEMG: Transmitting Muscle Effort to Vibration for Piano Pedagogy:

- Implemented real-time muscle-effort capture by integrating Delsys and BITalino EMG streams into a single calibration pipeline.
- Developed the MappEMG system to map EMG data to smartphone vibration for teacher-student feedback.
- Led evaluation workshops in five music conservatories to assess its pedagogical impact with more than forty piano learners.

Computer Music Group, Carnegie Mellon University

Research Assistant, Advisor: Prof. Roger B. Dannenberg

Mar. 2021 - Mar. 2022

Remote

Music Computing and Expectation Modeling:

- Investigated predictive models for musical sequences in contemporary music.
- Enhanced expectation-network algorithms by replacing simple counting with gradient-descent-based learning.

Music X Lab, New York University Shanghai

Research Assistant, Advisor: Asst. Prof. Gus Xia

Mar. 2020 - Jan. 2022

Shanghai, China

Multimodal Vocal Tutoring System:

- Developed an interactive score that plots breathing traces alongside pitch notation so singers can align phrasing with airflow.
- Integrated airflow, chest-expansion, and audio sensors to detect breathing patterns in real time during lessons.
- Prototyped a wearable device to track dynamic breathing movements.

ACHIEVEMENT

Funding and Grants

- **Student Award** (Collaborator: Hanwen Zhang), CIRMMT, \$8,000 CAD. 2025
- **TCH Innovation in Haptics Research Grant**, IEEE Technical Committee on Haptics, \$ 2,300 USD. 2024
- **FRQSC Doctoral Scholarship**, Fonds de recherche du Québec, \$ 83,334 CAD (Top 1/21). 2024 - 2028
- **Student Engagement Fund Winner** (Collaborators: Alberto Acquilino, Theodora Nestorova), McGill University, \$1,500 CAD. 2024
- **Inter-centre Research Exchange Funding**, CIRMMT, \$3,000 CAD. 2023
- **Student Award** (Collaborator: Theodora Nestorova), CIRMMT, \$10,000 CAD. 2023

Research Awards

- **ISMIR WIMIR Accomodation & Registration Grants**, International Society for Music Information Retrieval Conference. 2025
- **Best Oral Presentation**, The 2nd International Electronic Conference on Actuator Technology (IECAT). 2024
- **Best Oral Presentation**, COBS Student Colloquium. 2024
- **Graduate Excellence Award**, McGill University. 2022 - 2025
- **Online Conference Award**, CIRMMT. 2023 - 2024
- **Graduate Research Enhancement and Travel (GREAT) Award**, McGill University. 2023
- **NIME Ableton Scholarship**, International Conference on New Interfaces for Musical Expression. 2022
- **ISMIR Registration Grant**, International Society for Music Information Retrieval Conference. 2021

Other Honors

- **Differential Fee Waiver for International Students**, McGill University. 2022 - 2023
- **Outstanding Graduate**, Shanghai Jiao Tong University (Top 1%). 2022
- **Outstanding Teaching Assistant Award**, Shanghai Jiao Tong University (Top 1%). 2020
- **Excellent Teaching Assistant Award**, Foundational General Education of Shanghai Jiao Tong University. 2020
- **Outstanding Graduate of Beijing**, Beijing University of Posts and Telecommunications (Top 3%). 2019
- **Honorable Prize**, International Mathematical Contest in Modeling. 2018
- **Hengtong Photoelectric Enterprise Scholarship**, Beijing University of Posts and Telecommunications. 2017
- **1st Prize Academic Excellence Award**, Beijing University of Posts and Telecommunications. 2016, 2018

TEACHING

McGill University

Co-instructor with Champ Darabundit

- MUMT 302 New Media Production 1 2025 Winter
- MUMT 302 New Media Production 1 2024 Winter

Guest Lecturer

- MUPP 693 Performance Practice Seminar: *Feel the Music: Haptics in Music Learning* 2026 Winter
- MUPD 235 Music as a Profession: *Musical Haptics and Musicians* 2025 Winter
- MUMT 620 Gestural Control of Sound Synthesis: *Designing a Multimodal Interface for Music Pedagogy* 2023 Fall

New York University

Guest Lecturer

- CSCI-SHU 188 Introduction to Computer Music: *How to Design a Singing Tutoring Interface* 2021 Fall
- CSCI-SHU 188 Introduction to Computer Music: *Design a Single Thread System — a Singing Tutoring UI Demo* 2020 Fall

Shanghai Jiao Tong University

Teaching Assistant

- Introduction to Computer Music 2021 Winter
- Symphony 2020 Fall
- Chinese and Western Music Theory 2020 Spring

MENTORING

- **Jiayi Liang (2025), MS in Innovation Design Engineering, Imperial College London:** Vocal Flux: Visualizing your Voice through Muscle Waves
- **Gabrielle Seymour (2023-2024), BS in Computer Science, McGill University:** Audio-Visual Interactive for Ambient Music Generation

Committee Leadership Roles

- IEEE Technical Committee on Haptics (TCH) Student Representative 2025 - 2026
- ACM International Conference on Tangible Embedded and Embodied Interaction (TEI) 2026 Registration Chair 2025 - 2026
- ACM TEI 2026 Associate Chair of Work-In-Progress Track 2025 - 2026
- CIRMMT Student Representative 2024 - 2026
- HCI Montreal Help Desk Co-organizer 2025 - Present
- CIRMMT Student Coordinator for Research Axis 3 2023 - 2024
- Shanghai Jiao Tong University (SJTU) Academic Affair Fellow 2021 - 2022

Academic Reviews

*Reviewed 40 Submissions. Years marked with * indicate Special Recognitions for Outstanding Reviews.*

- IEEE Transactions on Haptics (ToH), article. 2026
- NIME, Papers and Workshops. 2021-2026
- ACM CHI, papers and Late Breaking Works. 2025*, 2026*
- ACM Design Interactive Systems (DIS), Papers. 2025
- ACM TEI, Papers and Pictorials. 2025, 2026
- ACM Creativity & Cognition (C&C), Papers. 2025
- ACM Ubiquitous Computing (UbiComp-ISWC), Demos. 2024
- EAI ArtsIT, Interactivity & Game Creation (ArtsIT), Papers. 2024

Volunteer

- Student Volunteer Lead of ISMIR 2026 2025 - 2026
- Volunteer mentor of HCI Montreal Help Desk 2024 - Present
- Volunteer of ISMIR 2024 2024
- Senior volunteer of NIME 2021 2021

TALKS AND PRESENTATIONS

- Music Learning with Haptics: Internal Awareness, Motor Control, and Collaborative Metaphor, *CIRMMT Haptics Workshop*. Apr. 2026
- The Duality of Breath: From Physiological Spontaneity to Conscious Control in Musical Expression, *McGill Doctoral Colloquium*. Apr. 2026
- How does breathing convey expressive and musical dynamics in piano performances, *Yamaha Corporation*. Dec. 2025
- Touching the Unseen: Enhancing Music Learning with Wearable Haptics, *Yamaha Corporation*. Sep. 2025
- ScapuLearn: Minimally-intrusive Estimation of Finger and Postural Dynamics via Scapular Musculoskeletal Sensing, *Sony CSL*. Aug. 2025
- Bridging Kinesthetic Mismatches in Music Learning with Wearable Haptic Devices, *COBS Student Colloquium*. May. 2025
- CantusPlay: An Interactive Singing Interface for Practicing Improvised Renaissance Counterpoint, *COBS Student Colloquium*. May. 2025
- Exploring on Breathing Challenges and Technology Assisted Solutions, *The Voice Foundation Annual Symposium*. May. 2025
- How to write a success CIRMMT student award proposal, *CIRMMT Colloquial Colloquium*. Jan. 2025
- Assessing the Impact of Force Feedback in Musical Knobs on Performance and User Experience, *IECAT Conference 2024*. Nov. 2024
- SenSing: Transmitting Kinesthetic Breathing Experiences for Voice Pedagogy, *PAVA Symposium 2024*. Oct. 2024
- Preliminary Exploration of Kinesthetic Breathing Feedback for Enhanced Vocal Pedagogy, *COBS Student Colloquium*. May 2024
- Designing a Multimodal Interface for Music Pedagogy, *CIRMMT Workshop with Prof. Karon MacLean*. Nov. 2023
- SenSing: Transmitting Kinesthetic Breathing Experiences for Voice Pedagogy, *CIRMMT MMR Re-Open Celebration*. Oct. 2023
- SenSing: Transmitting Kinesthetic Breathing Experiences for Voice Pedagogy, *COBS Student Colloquium*. May 2023
- MappEMG – Feeling the muscle activity of classical musicians, *CIRMMT Workshop on Piano Knowledge Transfer*. May 2023
- Sensing the Singing Body, *CIRMMT Workshop with Pamela Z*. Jan. 2023
- Sensing the Breath: Multimodal Singing Tutoring Interfaces with Breath Guidance, *Music + AI Reading Group, Mila*. Oct. 2022
- Multimodal Singing Tutoring Interfaces Design for Breathing Guidance, *IDMIL, McGill University*. Sep. 2022

ACTIVITIES

- **Soprano Classical Singer:** Under the guidance of Dr. Irene M Feher and Giancarlo Scalia. 2022 - Present
- **Choir Singer:** Served as an alto singer in the Shanghai Jiao Tong University (SJTU) Chamber Choir. 2020 - 2021
- **Orchestra Guzheng Player:** Played the Guzheng (Chinese zither) in the SJTU Rongchang Chinese and Western Orchestra. 2019 - 2020
- **Designer:** Designed the graduation souvenir shirt for the economics double major Class at Peking University. 2019
- **Field Research Leader:** Led the Henan No.10 team during the Peking University Enterprise Survey for Entrepreneurship. 2018
- **Art Director:** Managed the organization of singing and dancing performances at School of Digital Media. 2015 - 2019

SKILLS

- **Programming:** Python*, C++, Javascript*.
- **Audio Production:** Logic Pro*, Audition, PureData*, MAX MSP
- **Prototyping:** Arduino*, Fusion 360, 3D Printing.
- **VR, AR and Game:** Maya*, Processing*, Unity, After Effects.
- **Graphic Design:** Illustrator*, Photoshop*.
- **Toolkits:** scikit, PyTorch.
- **Languages:** Mandarin (native), English (advanced), French (beginner).

*Items marked with *: professional proficiency; others: used at least once in projects.*

REFERENCES

Prof. Marcelo M. Wanderley	McGill University	marcelo.wanderley@mcgill.ca
Assoc. Prof. Isabelle Cossette	McGill University	isabelle.cossette1@mcgill.ca
Prof. Roger B. Dannenberg	Carnegie Mellon University	rbd@cs.cmu.edu
Asst. Prof. Gus Xia	MBZUAI	gus.xia@mbzuai.ac.ae
Dr. Akira Maezawa	Yamaha Corporation	akira.maezawa@music.yamaha.com

Available upon request.